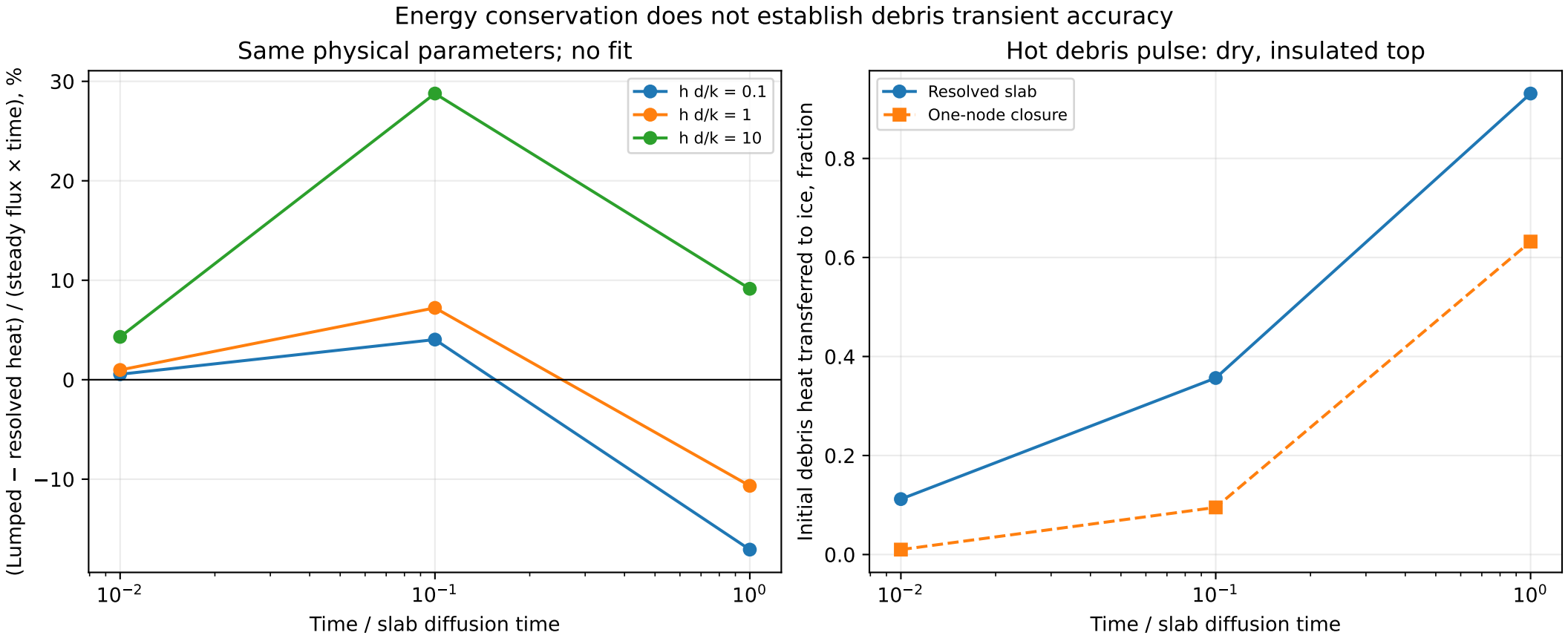
Energy conservation does not establish debris transient accuracy
Same physical parameters; no fit
Hot debris pulse: dry, insulated top
(Lumped − resolved heat) / (steady flux × time), %
Initial debris heat transferred to ice, fraction
Time / slab diffusion time
Time / slab diffusion time
h d/k = 0.1
h d/k = 1
h d/k = 10
Resolved slab
One-node closure
30
20
10
0
−10
0.8
0.6
0.4
0.2
0.0
10^{-2}
10^{-1}
10^{0}
10^{-2}
10^{-1}
10^{0}